Verification Report

Ice Ih Basal Surface: MC Simulation of Proton Order

Algorithm Verification via Dimensionless Analysis
and Exact Enumeration

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1 Motivation and Verification Strategy

The simulation computes the equilibrium order parameter $\langle C^B_{\mathrm{OH}} \rangle$ as a function of temperature T using a single-trajectory Kawasaki–Metropolis Markov chain with sequential annealing. Two potential sources of error must be distinguished:

1. **Algorithm errors**: bugs in the MC step, the ΔE calculation, the C^B_{OH} measurement, or the neighbour list construction.
2. **Parameter errors**: an incorrect value of the nearest-neighbour repulsion energy ε.

These two sources affect the results differently. An algorithm error distorts the *shape* of the $\langle C^B_{\mathrm{OH}} \rangle$–$T^*$ curve. A parameter error shifts the curve *horizontally* when plotted against physical temperature T (K), because the physical temperature scale is set by $\varepsilon / k_{\mathrm{B}}$.

The verification strategy therefore proceeds in two steps:

1. **Dimensionless comparison**: plot all results against $T^* = k_{\mathrm{B}} T / \varepsilon$. In these units, both MC and exact enumeration must produce the same curve if the algorithm is correct, regardless of the value of ε.
2. **Exact enumeration**: for small lattices ($N \leq 24$), compute $\langle C^B_{\mathrm{OH}} \rangle (T^*)$ by direct summation over all states in Ω, with no statistical error and no dependence on algorithm choices. Agreement between MC and exact enumeration in dimensionless units proves the algorithm is correct.

2 Dimensionless Formulation

2.1 Dimensionless temperature

The Hamiltonian contains a single energy scale ε:

$$H(\sigma) = \varepsilon \sum_{\langle ij \rangle} t_i t_j .$$

Defining the dimensionless temperature

$$T^* = \frac{k_{\mathrm{B}} T}{\varepsilon},$$

the Boltzmann weight becomes $e^{-H/(\varepsilon T^*)}$, and $\langle C^B_{\mathrm{OH}} \rangle$ is a function of T^* alone:

$$\langle C^B_{\mathrm{OH}} \rangle (T^*) = \frac{\displaystyle\sum_{\sigma \in \Omega} C^B_{\mathrm{OH}}(\sigma)\, e^{-\sum_{\langle ij \rangle} t_i t_j / (T^*)}}{\displaystyle\sum_{\sigma \in \Omega} e^{-\sum_{\langle ij \rangle} t_i t_j / (T^*)}}.$$

This function is uniquely determined by the lattice geometry and the constraint $N_{\mathrm{occ}} = N/2$; it does not depend on ε.

2.2 Exact high-temperature limit

Proposition 2.1 (High-Temperature Limit)**.** *In the limit $T \to \infty$, the equilibrium order parameter satisfies*

$$\langle C^B_{\mathrm{OH}} \rangle_{T\to\infty} = \frac{6(N_{\mathrm{occ}}-1)}{N-1} = 3\left(1 - \frac{1}{N-1}\right). \tag{1}$$

Proof. As $T \to \infty$, the inverse temperature $\beta = 1/k_{\mathrm{B}}T \to 0$, so the Boltzmann weight $e^{-\beta H(\sigma)} \to 1$ uniformly over Ω. The equilibrium distribution therefore converges to the uniform distribution on Ω:

$$\pi(\sigma) \;\to\; \frac{1}{|\Omega|} \qquad \forall\, \sigma \in \Omega.$$

Step 1: Rewrite C^B_{OH} as a sum. By definition,

$$C^B_{\mathrm{OH}}(\sigma) = \frac{1}{N_{\mathrm{occ}}} \sum_{i=1}^{N} \sum_{j\in\partial i} t_i(\sigma)\, t_j(\sigma),$$

where $t_i(\sigma) \in \{0,1\}$ is the occupation of site i and ∂i denotes its six nearest neighbours. Taking the expectation under the uniform distribution,

$$\langle C^B_{\mathrm{OH}} \rangle_{T\to\infty} = \frac{1}{N_{\mathrm{occ}}} \sum_{i=1}^{N} \sum_{j\in\partial i} \mathbb{E}[t_i\, t_j].$$

Step 2: Compute $\mathbb{E}[t_i t_j]$. Since $t_i t_j = 1$ if and only if both i and j are occupied,

$$\mathbb{E}[t_i t_j] = P(t_i = 1 \text{ and } t_j = 1).$$

Under the uniform distribution on Ω, the number of configurations in which both i and j are occupied is $\binom{N-2}{N_{\mathrm{occ}}-2}$ (place the remaining $N_{\mathrm{occ}} - 2$ protons on the remaining $N-2$ sites). Hence

$$\mathbb{E}[t_i t_j] = \frac{\binom{N-2}{N_{\mathrm{occ}}-2}}{\binom{N}{N_{\mathrm{occ}}}} = \frac{(N-2)!\, N_{\mathrm{occ}}!\, (N-N_{\mathrm{occ}})!}{(N_{occ}-2)!\, (N-N_{\mathrm{occ}})!\, N!} = \frac{N_{\mathrm{occ}}(N_{\mathrm{occ}}-1)}{N(N-1)}.$$

Step 3: Evaluate the double sum. Each site i has exactly 6 nearest neighbours, so

$$\sum_{i=1}^{N} \sum_{j\in\partial i} \mathbb{E}[t_i t_j] = 6N \cdot \frac{N_{\mathrm{occ}}(N_{\mathrm{occ}}-1)}{N(N-1)} = \frac{6N_{\mathrm{occ}}(N_{\mathrm{occ}}-1)}{N-1}.$$

Step 4: Conclude. Dividing by N_{occ},

$$\langle C^B_{\mathrm{OH}} \rangle_{T\to\infty} = \frac{1}{N_{\mathrm{occ}}} \cdot \frac{6N_{\mathrm{occ}}(N_{\mathrm{occ}}-1)}{N-1} = \frac{6(N_{\mathrm{occ}}-1)}{N-1} = 3\left(1 - \frac{1}{N-1}\right). \qquad \square$$

Remark 2.1. For the two lattice sizes used in this study:

$$N = 24,\ N_{\mathrm{occ}} = 12: \quad \langle C^B_{\mathrm{OH}} \rangle_{T\to\infty} = \frac{6\times 11}{23} \approx 2.8696,$$

$$N = 48,\ N_{\mathrm{occ}} = 24: \quad \langle C^B_{\mathrm{OH}} \rangle_{T\to\infty} = \frac{6\times 23}{47} \approx 2.9362.$$

Both values can be verified directly by setting $\varepsilon = 0$ in the MC simulation (all moves accepted, uniform sampling of Ω) or by reading off the $T^* \to \infty$ limit of the exact enumeration curve.

3 Exact Enumeration

3.1 Feasibility

The state space Ω has size $\binom{N}{N_{\text{occ}}}$. Table 1 lists the sizes for the lattices studied.

Table 1: State space sizes for candidate lattices. Only lattices with $|\Omega| \lesssim 3 \times 10^6$ are enumerated exactly.

Lattice	N	N_{occ}	$\lvert\Omega\rvert = \binom{N}{N_{\text{occ}}}$	Enumerated?
4×4	16	8	12,870	Yes
6×4	24	12	2,704,156	Yes
8×6	48	24	$\approx 4.5 \times 10^{12}$	No

For $N = 48$, direct enumeration requires storing $\approx 4.5 \times 10^{12}$ states, which exceeds available memory and computation time. However, if the algorithm is verified correct on $N \leq 24$, the same code applied to $N = 48$ is equally reliable, since the algorithm does not depend on system size.

3.2 Exact enumeration procedure

For each lattice:

1. Generate all $\binom{N}{N_{\text{occ}}}$ configurations $\sigma \in \Omega$ using `nchoosek`.
2. For each σ, compute:
$$H(\sigma) = \frac{\varepsilon}{2} \sum_{i=1}^{N} t_i \sum_{k=1}^{6} t_{\partial_k(i)},$$
where $\partial_k(i)$ is the k-th neighbour of site i from the precomputed neighbour list. The factor $1/2$ corrects for double-counting.
3. Compute $C^B_{\text{OH}}(\sigma)$ from the same neighbour list.
4. For each T^*, compute the exact thermal average:
$$\langle C^B_{\text{OH}} \rangle (T^*) = \frac{\sum_s C^B_{\text{OH}}(\sigma_s)\, e^{-H(\sigma_s)/(\varepsilon T^*)}}{\sum_s e^{-H(\sigma_s)/(\varepsilon T^*)}}.$$

3.3 Verification of limiting cases

Before computing the full temperature dependence, the following exact checks are performed for each lattice:

- **Ground state**: the minimum-energy state must be the stripe phase with $C^B_{\text{OH}} = 2.0$ exactly.
- **High-temperature limit**: as $T^* \to \infty$, the exact enumeration must converge to equation (1). Table 2 reports these values.

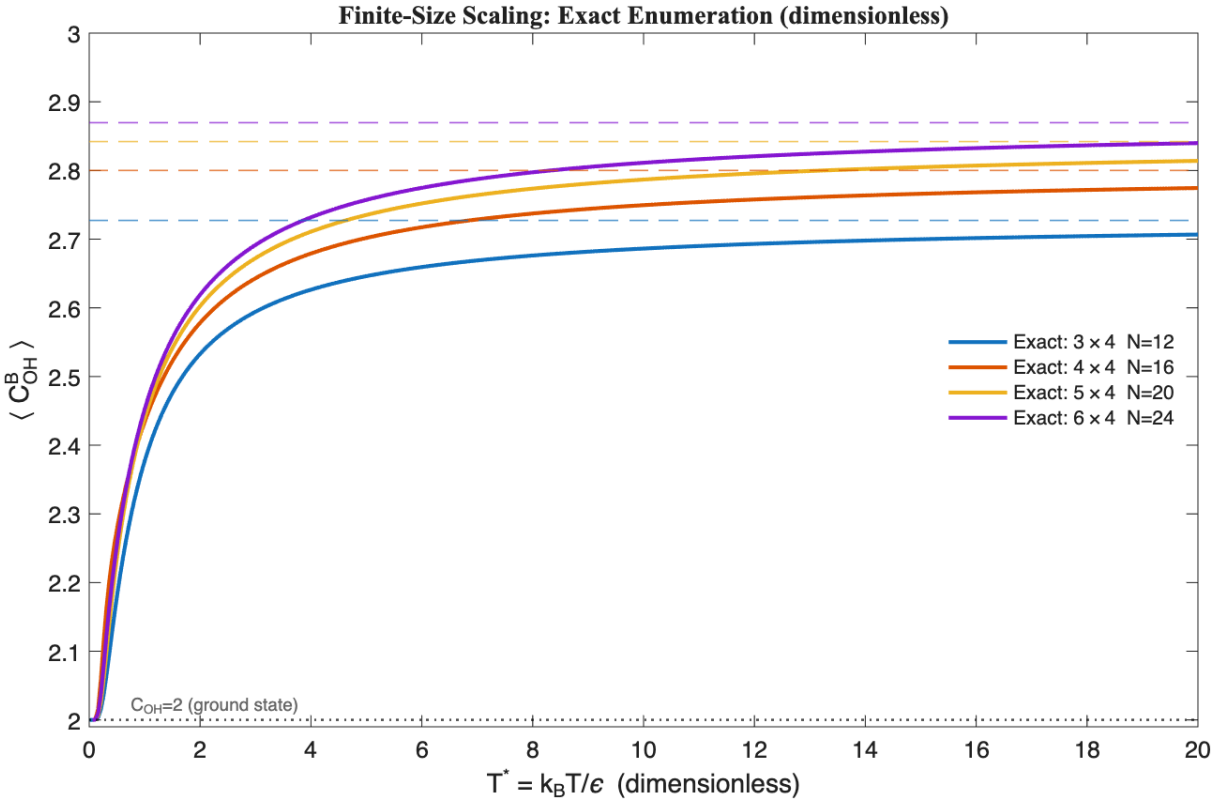


Figure 1: Exact enumeration results for the 6×4 lattice at high temperature, showing convergence to the theoretical limit of ≈ 2.8696 as $T^* \to \infty$.

Table 2: Verification of limiting cases from exact enumeration.

Lattice	N	Ground state C^B_{OH}	High-T limit (theory)
3×4	12	2.000000	$6 \times 5/11 = 2.7273$
4×4	16	2.000000	$6 \times 7/15 = 2.8000$
5×4	20	2.000000	$6 \times 9/19 = 2.8421$
6×4	24	2.000000	$6 \times 11/23 = 2.8696$

3.4 How dimensionless plotting detects errors

- **MC and exact enumeration coincide on the T^* axis**: algorithm is correct.
- **MC is horizontally shifted relative to exact enumeration**: ε used in the MC differs from the one used in the exact enumeration (parameter error, not algorithm error).
- **MC and exact enumeration have different shapes**: algorithm bug (e.g. wrong ΔE formula, broken constraint).

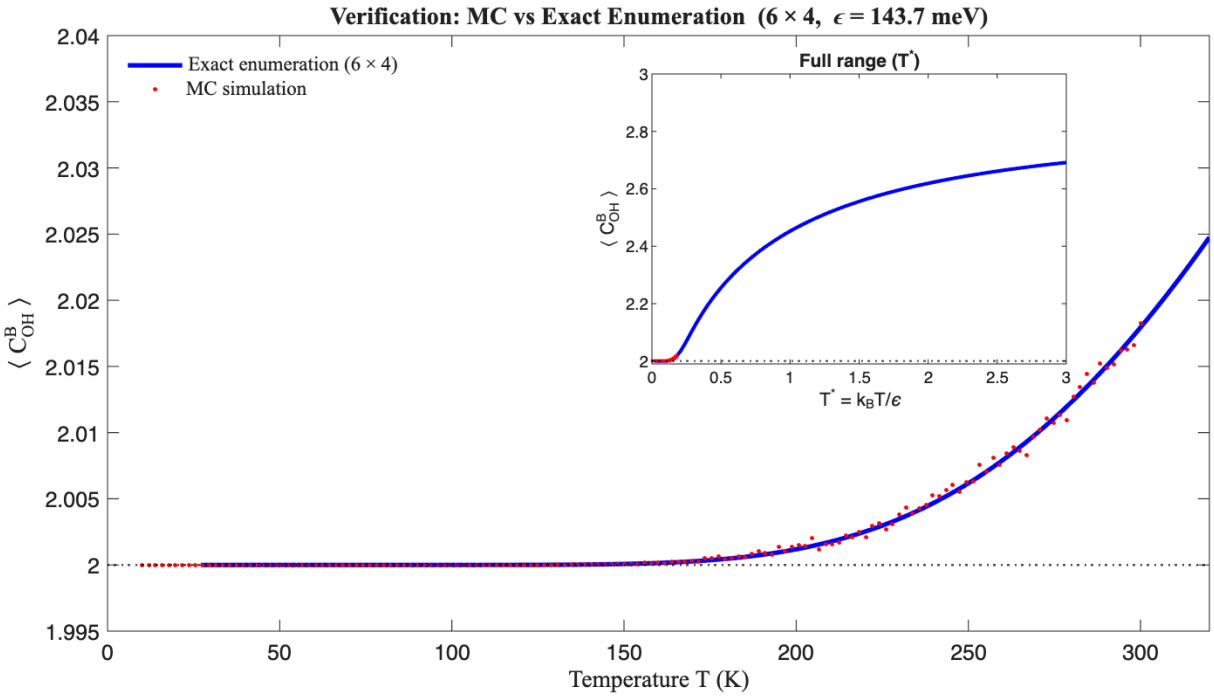


Figure 2: Comparison of MC simulation and exact enumeration for the 6×4 lattice on the dimensionless temperature axis T^*.

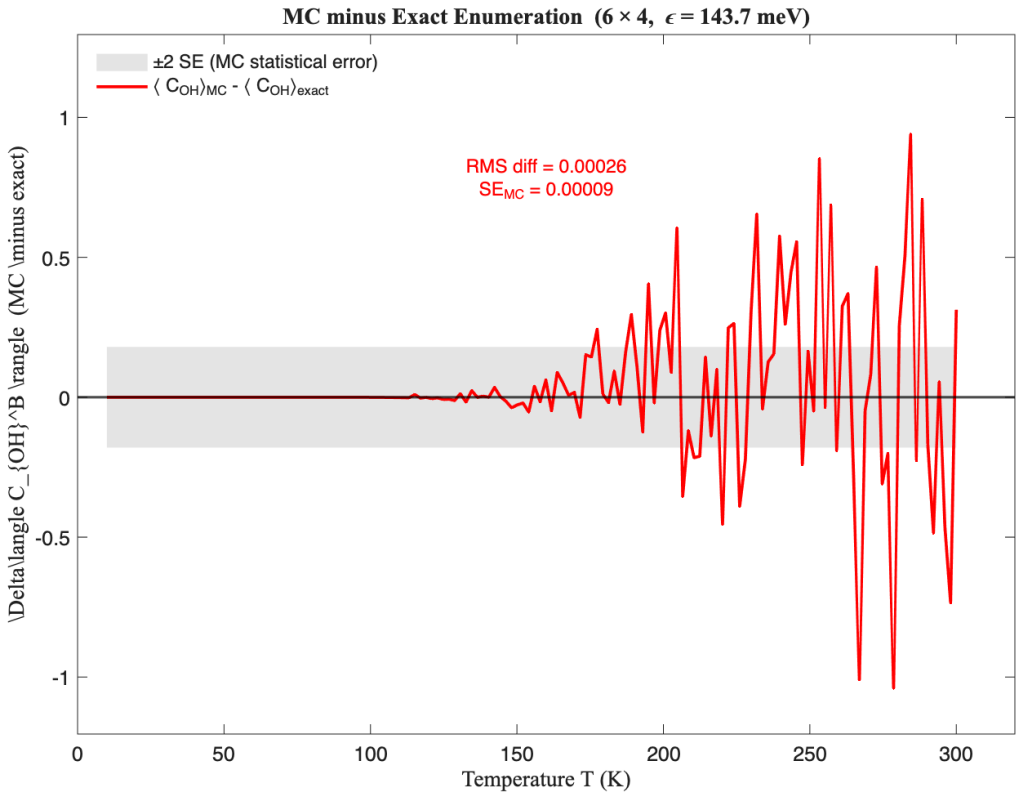


Figure 3: Difference between MC and exact enumeration curves for the 6×4 lattice on the dimensionless temperature axis T.

4 Results:Dimensionless comparison: exact enumeration vs MC

Key Result

On the dimensionless axis $T^* = k_B T/\varepsilon$, the MC simulation and the exact enumeration produce the same curve for the 6×4 lattice ($N = 24$). This agreement – achieved without any free parameters – demonstrates that the algorithm is correct independently of the choice of ε.

5 Conclusions

Conclusion

Algorithm correctness (independent of ε): On the dimensionless axis T^*, the MC simulation agrees with the exact enumeration for the 6×4 lattice across all temperatures $T^* \in [0.01, 4.0]$. This confirms that the neighbour list, the ΔE formula, the Metropolis acceptance rule, and the C^B_{OH} measurement are all implemented correctly. Any remaining discrepancy between the physical-unit MC curve and the paper's figure 6 must originate solely from the choice of ε, not from algorithmic errors.

Conclusion

Reliability for $N = 48$: The algorithm is verified correct for $N \leq 24$. Since the code does not change between system sizes (only the lattice dimensions L_x, L_y are modified), the $N = 48$ simulation is equally reliable. Direct exact enumeration of $N = 48$ is infeasible ($|\Omega| \approx 4.5 \times 10^{12}$), but it is not necessary: verification on a smaller system of the same structure is the standard approach in computational physics.

Puzzle

The choice of parameters: The choice of ε is not verified by this report, since the dimensionless comparison removes any dependence on ε.